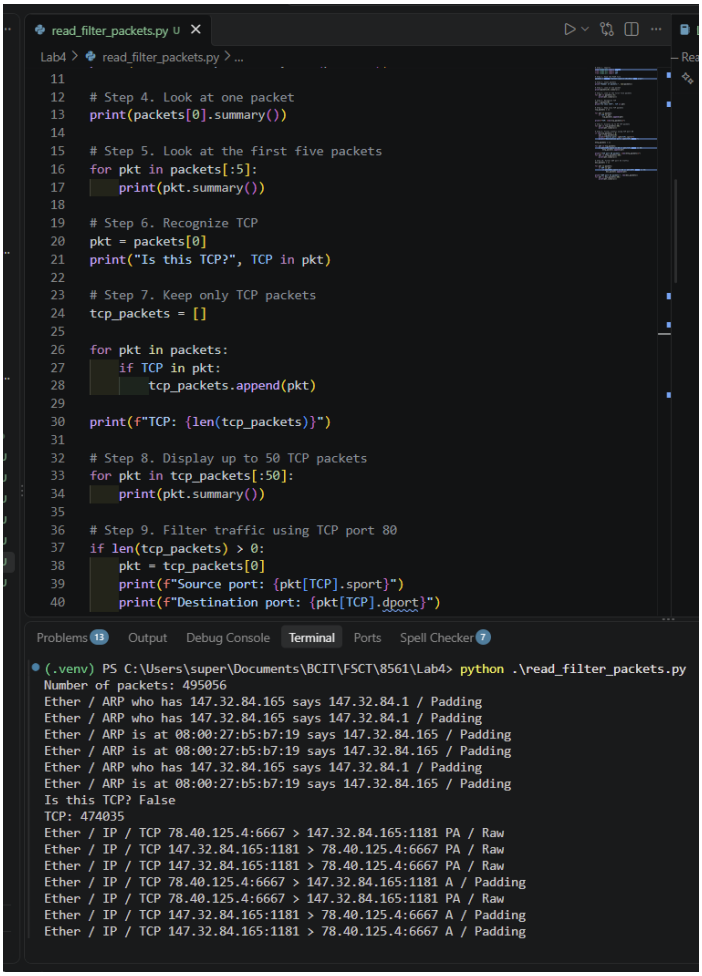
Braeden Venne

Lab 4: Network Traffic Analysis Using Scapy

Part 1 – Read and Filter Packets



1. How many TCP, TCP-port-80, and UDP-port-53 packets did you find?
    - TCP: 474035
    - TCP-port-80: 22430
    - UDP-port-53: 57
2. Why might a filter return zero packets?
    - The PCAP file may not contain that specific protocol or port traffic you are trying to filter for
    - The filter condition you set may have an error or incorrect code logic
3. What is the difference between filtering by protocol and filtering by port?
    - Filtering by protocol filters by transport layer type like TCP or UDP
    - Filtering by port filters by specific application services such as Port 80 for HTTP or Port 53 for DNS

Braeden Venne

Part 2 – Inspect Packet Headers

```
(.venv) PS C:\Users\super\Documents\BCIT\FSCT\8561\Lab4> python .\inspect-packet-headers.py
###[ Ethernet ]###
  dst       = 08:00:27:b5:b7:19
  src       = 00:1e:49:db:19:c3
  type      = ARP
###[ ARP ]###
     hwtype    = Ethernet (10Mb)
     ptype     = IPv4
     hwlen     = 6
     plen      = 4
     op        = who-has
     hwsrc     = 00:1e:49:db:19:c3
     psrc      = 147.32.84.1
     hwdst     = 08:00:27:b5:b7:19
     pdst      = 147.32.84.165
###[ Padding ]###
        load      = b'\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'

Contains IPv4? False
78.40.125.4 -> 147.32.84.165
147.32.84.165 -> 78.40.125.4
147.32.84.165 -> 78.40.125.4
78.40.125.4 -> 147.32.84.165
78.40.125.4 -> 147.32.84.165
TCP flags: PA
Size (bytes): 60
Timestamp: 1313153171.504306
Source: 78.40.125.4 Destination: 147.32.84.165 Bytes: 83 Time: 1313153244.357711
Source: 147.32.84.165 Destination: 78.40.125.4 Bytes: 83 Time: 1313153244.358073
Source: 147.32.84.165 Destination: 78.40.125.4 Bytes: 83 Time: 1313153244.358085
Source: 78.40.125.4 Destination: 147.32.84.165 Bytes: 60 Time: 1313153244.388656
Source: 78.40.125.4 Destination: 147.32.84.165 Bytes: 120 Time: 1313153273.62872
```

1. What do source and destination IP addresses tell you?
    - The source address tells you who sent the packet and the destination address tells you who received it.
2. What is the purpose of TCP flags?
    - TCP flags control the connection status, like starting the communication (SYN) or ending (FIN) it.
3. Why are timestamps useful when investigating possible flooding?
    - Timestamps are useful when investigating potential flooding because it shows you exactly how many packets arrived per second to detect sudden traffic spikes.

Braeden Venne

Part 3 - Inspect payloads and possible information exposure

```
(.venv) PS C:\Users\super\Documents\BCIT\FSCT\8561\Lab4>  python .\inspect-payloads.py
Contains Raw payload? False
No Raw payload in this packet
Payload sample: b'PING :barjavel.freenode.net\r\n'
PING :barjavel.freenode.net

Possible plaintext HTTP header found
(.venv) PS C:\Users\super\Documents\BCIT\FSCT\8561\Lab4>
```

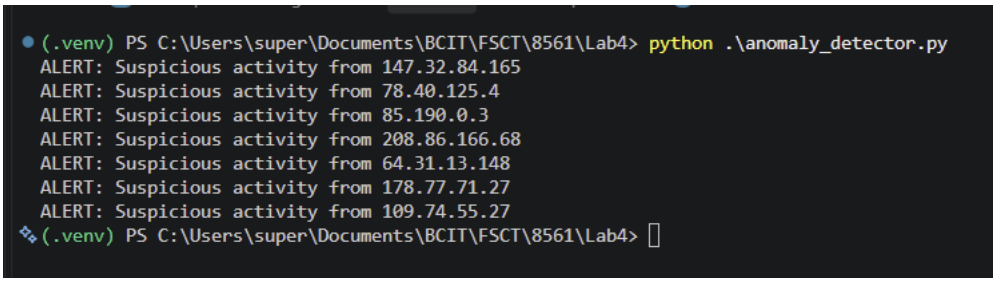
1. Did you find readable payload text?
    - Yes, found readable plaintext IRC commands like “PING :barjavel.freenode.net”
2. What might plaintext HTTP expose?
    - HTTP plaintext may expose user sensitive information such as their passwords, cookies, email address, etc.
3. Why can you not inspect the contents of encrypted HTTPS traffic using these simple commands?
    - HTTPS encrypts the payload into random bytes, so you cannot read it without the decryption key.

Part 4 – Build a Basic Traffic Analyzer

```
(.venv) PS C:\Users\super\Documents\BCIT\FSCT\8561\Lab4> python .\traffic_sniffer.py
Source: 78.40.125.4     Destination: 147.32.84.165      Protocol: TCP   Source Port: 6667       Destination Port: 1181
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 78.40.125.4     Destination: 147.32.84.165      Protocol: TCP   Source Port: 6667       Destination Port: 1181
Source: 78.40.125.4     Destination: 147.32.84.165      Protocol: TCP   Source Port: 6667       Destination Port: 1181
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 78.40.125.4     Destination: 147.32.84.165      Protocol: TCP   Source Port: 6667       Destination Port: 1181
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 78.40.125.4     Destination: 147.32.84.165      Protocol: TCP   Source Port: 6667       Destination Port: 1181
Source: 78.40.125.4     Destination: 147.32.84.165      Protocol: TCP   Source Port: 6667       Destination Port: 1181
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 78.40.125.4     Destination: 147.32.84.165      Protocol: TCP   Source Port: 6667       Destination Port: 1181
Source: 78.40.125.4     Destination: 147.32.84.165      Protocol: TCP   Source Port: 6667       Destination Port: 1181
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
Source: 147.32.84.165   Destination: 78.40.125.4        Protocol: TCP   Source Port: 1181       Destination Port: 6667
TCP packets: 474035
UDP packets: 2386
```

Braeden Venne

Part 5 - Build a simple PCAP-based IDS



Part 6 – Reflection Questions

1. Why is packet inspection important for network security?
    - It allows you to see network activity, detect any suspicious behavior, and find malicious commands.
2. What types of security risks can arise from unencrypted traffic?
    - Unencrypted traffic allows for eavesdroppers to steal sensitive information such as plaintext HTTP cookies, password, or user agent details using simple packet capturing tools such as tcpdump or Wireshark.
3. What are the limitations of passive packet sniffing and PCAP analysis?
    - It only logs historical traffic without stopping live attacks and it cannot view contents encrypted with things like HTTPS or TLS.
4. How does encryption help prevent information leakage?
    - It turns the packets into unreadable ciphertext so that sniffing tools cannot read the data contained within.
5. Why might a simple packet-rate detector produce false positives?
    - Simple packet-rate detectors may struggle with legitimate high-volume traffic such as large file downloads or multi-user network routers triggering the set threshold.

Braeden Venne

Part 7 – Security Analysis

Inspection of provided PCAP file revealed continuous communication between internal host 147.32.84.165 and an external IP 78.40.125.4. Transport layer filtering identified 474,035 TCP packets alongside 2,386 UDP packets. Payload inspection of the raw packets uncovered unencrypted IRC communication, displaying text strings such as PING :barjavel.freenode.net. Running the Python anomaly detector flagged multiple source IPs such as 147.32.84.165 and 78.40.125.4 sending bursts exceeding 20 packets within the set 5-second windows.

One of the main threats shown in these observations is the unencrypted data exposure, since plain text IRC commands and headers can easily be captured by packet sniffers on the network. Additionally, the high packet volume exceeding 20 packets in 5 seconds from specific IPs creates a risk of network congestion or service disruption. Finally, seeing the IRC protocol traffic on standard ports highlights unauthorized application activity running on internal endpoints.

To reduce network exposure all web and application traffic should be enforcing TLS and HTTPS encryption so sensitive headers and data are not visible in plain text. Additionally, firewalls should be blocking unencrypted IRC ports. Finally, the host should be disconnected from the network and inspected to find and remove any unauthorized background applications.

This lab highlighted a few key limitations of packet capture and threshold-based detection. PCAP analysis only logs historical traffic, so it cannot be used to stop ongoing or active attacks. Additionally, with PCAP analysis you cannot view encrypted traffic such as TLS or HTTPS. Lastly, simple packet-rate detectors may struggle with legitimate high-volume traffic such as large file downloads or multi-user network routers triggering the set threshold.